

Two Questions from Fractional Adams–Moser–Trudinger Inequalities Were Answered by Hyder

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Lane 0 — literature crosswalk

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Status. Literature already answered. Hyder’s Theorem 1.1 gives a positive answer to Martinazzi’s stronger critical-sharpness question, under a strictly stronger norm constraint. Hyder’s Theorem 1.3 also settles the highlighted odd-dimensional model equation. The two other questions on the same source page have later partial answers but are not claimed fully resolved here.

1 Exact source questions

Martinazzi proves the sharp critical exponential inequality for $\tilde{H}^{n/p,p}(\Omega)$ on finite-measure domains. He then asks for the stronger endpoint divergence

$$\sup_{\substack{u \in \tilde{H}^{n/p,p}(\Omega) \\ \|(-\Delta)^{n/(2p)}u\|_{L^p(\Omega)} \leq 1}} \int_{\Omega} f(|u|) e^{\alpha_{n,p}|u|^{p'}} dx = \infty \quad (1)$$

for every nonnegative f tending to infinity. Immediately before the open questions section, he also identifies the critical equation $(-\Delta)^{n/2}u = f(u)$, especially $f(u) = ue^{u^2}$, as open in odd dimensions $n \geq 3$.

so that

$$F(u) \geq \left(\Lambda - \frac{n^2 p'}{4\alpha_{n,2}} \right) \|u\|^2 - \tilde{C},$$

i.e. F is well-defined, bounded from below and coercive on $\tilde{H}^{\frac{n}{2},2}(\Omega)$. A minimizing sequence u_k is therefore bounded in $\tilde{H}^{\frac{n}{2},2}(\Omega)$, hence weakly converging to a minimizer $u_0 \in A$. By taking first variations (the set A is open) it follows that

$$(-\Delta)^{\frac{n}{2}} u_0 = \frac{nK e^{nu_0}}{2\Lambda \int_{\Omega} K(e^{nu_0} - 1) dx},$$

and up to adding a constant we find a solution to (26). $\square$

One can also prove existence results for more general semilinear equations, say

$$(-\Delta)^{\frac{n}{2}} u = f(u) \quad \text{in } \Omega \quad (27)$$

with $u \in \tilde{H}^{\frac{n}{2},2}(\Omega)$ and f critical or subcritical, in the spirit for instance of the works of Adimurthi [4] and Iannizzotto and Squassina [23], even in the case of the fractional p -Laplacian, but we will not do that. We only remark that in the case when f is critical, e.g. $f(u) = u e^{u^2}$, a crucial ingredient is (8) for $p = a = 2$, which is known only in dimension 1 (by [24]) and in even dimension (by [2]), hence the critical case in odd dimension ≥ 3 is open. In arbitrary even dimension we mention the work of Lakkis [29]. The subcritical case should instead present no major difficulties since the functional corresponding to (27) should satisfy the Palais-Smale condition. For the regularity theory of nonlinear nonlocal equations we refer the reader e.g. to [27], [28], [42] and [44].

4 Open questions

An interesting question is whether fractional Adams-Moser-Trudinger inequalities hold for some domains of infinite measure, in the spirit of the results of Gianni Mancini-Sandeep [34] and Battaglia-Gabriele Mancini [6] who in dimension 2 and in the classical case $p = 2$ proved that the inequality

$$\sup_{u \in C_c^\infty(\Omega): \|\nabla u\|_{L^2} \leq 1} \int_{\Omega} (e^{4\pi u^2} - 1) dx < \infty \quad (28)$$

holds if and only if $\lambda_1(\Omega) > 0$, where $\lambda_1(\Omega)$ is the first eigenvalue of the Laplace operator with Dirichlet boundary conditions on Ω (open set in $\mathbb{R}^2$).

Another natural question is whether one can replace the spaces $\tilde{H}^{\frac{n}{p},p}(\Omega)$ with the spaces $\tilde{W}^{\frac{n}{p},p}(\Omega)$, defined via a double integral (see e.g. (29) below). This appears to be unknown already in dimension 1, except when $p = 2$ (see Proposition 15).

As already discussed in the introduction it would be interesting to prove the sharpness of the constants in the stronger form

$$\sup_{u \in \tilde{H}^{\frac{n}{p},p}(\Omega), \|(-\Delta)^{\frac{n}{2p}} u\|_{L^p(\Omega)} \leq 1} \int_{\Omega} f(|u|) e^{\alpha_{n,p}|u|^{p'}} dx = \infty$$

for any function

$$f : [0, \infty) \rightarrow [0, \infty) \quad \text{with} \quad \lim_{t \rightarrow \infty} f(t) = \infty,$$

as already known in the non-fractional case and in dimension 1.

Figure 1: Compiled source page 13: the odd-dimensional critical case and all three questions in Section 4.

2 The theorem-level answer

Hyder's paper explicitly repeats (1), calls it Martinazzi's question, and proves the following theorem.

Hyder, Theorem 1.1. For every finite-measure open $\Omega \subset \mathbb{R}^n$, $1 < p < \infty$, and Borel $f \geq 0$ with $f(t) \rightarrow \infty$,

$$\sup_{u \in \tilde{H}^{n/p,p}(\Omega)} \int_{\Omega} f(|u|) e^{\alpha_{n,p}|u|^{p'}} dx = \infty. \quad (2)$$

$$\|u\|_{L^p(\Omega)}^p + \|(-\Delta)^{n/(2p)} u\|_{L^p(\mathbb{R}^n)}^p \leq 1$$

This is stronger than the requested statement. Every function admissible in (2) satisfies $\|(-\Delta)^{n/(2p)}u\|_{L^p(\Omega)} \leq 1$, so its admissible set is contained in that of (1). Divergence over the smaller set forces divergence over the larger one. The constant is exactly Martinazzi's $\alpha_{n,p}$ and the quantifier over slowly growing f is unchanged.

Hyder constructs smooth compactly supported fractional Moser functions and proves the precise estimate

$$\|(-\Delta)^{n/(2p)}u_\varepsilon\|_{L^p(\mathbb{R}^n)} \leq \left(1 + \frac{C}{\log(1/\varepsilon)}\right)^{1/p}.$$

The $O(1/\log)$ error is the missing precision: after normalization it leaves the critical exponential mass bounded below while $|u_\varepsilon| \rightarrow \infty$ on a shrinking ball, so any factor $f(|u_\varepsilon|) \rightarrow \infty$ forces divergence.

Hyder, Theorem 1.3. For every finite-measure open $\Omega \subset \mathbb{R}^n$, every $0 < \lambda < \lambda_1$, and every $b > 0$, there is a nontrivial weak solution of

$$(-\Delta)^{n/2}u = \lambda u e^{bu^2} \quad \text{in } \Omega, \quad u \in \tilde{H}^{n/2,2}(\Omega).$$

The theorem has no parity restriction, hence it contains every odd $n \geq 3$ highlighted by Martinazzi. It resolves the displayed model critical nonlinearity, not every conceivable critical f .

3 Crosswalk and residual scope

Source item	Later theorem	Exact conclusion
Strong sharpness for every $f(t) \rightarrow \infty$	Hyder 1.1	Fully answered yes, even under the full $H^{n/p,p}$ norm.
Odd-dimensional ue^{u^2} equation	Hyder 1.3	Nontrivial weak solution in all dimensions for $0 < \lambda < \lambda_1$, $b > 0$.
Replace Bessel by Slobodeckij spaces	Parini–Ruf 1.1; Sk 1.1	Positive inequalities and a domain criterion exist for $0 < s < 1$, $sp = n$, but the exact optimal exponent was still stated unknown in 2021.
Infinite-measure domains	Sk 1.1	For the Slobodeckij/truncated-exponential form, validity is equivalent to a fractional Poincaré inequality. This does not by itself settle the literal Bessel-potential, derivative-only version.

Thus no new-proof target remains for the two selected claims. The exact Slobodeckij sharp constant is a separate viable problem, but it is not silently promoted to a resolution here.

4 Verification

The source statement was checked in the compiled source PDF, page 13. Hyder's question identification and Theorem 1.1 were checked on official arXiv PDF page 3; Theorem 1.3 was checked on page 5. The implication from the stronger full-norm admissible class to Martinazzi's class is immediate set inclusion. The later-status statements were checked against Parini–Ruf [arXiv:1607.07681](#) and Sk [arXiv:2002.11747](#), especially Sk's Theorem 1.1 and its explicit statement that the optimal Slobodeckij constant remained unknown.

References

- [1] L. Martinazzi, *Fractional Adams–Moser–Trudinger type inequalities*, Nonlinear Analysis 127 (2015), 263–278, [arXiv:1506.00489](#).
- [2] A. Hyder, *Moser functions and fractional Moser–Trudinger type inequalities*, Nonlinear Analysis 146 (2016), 185–210, [arXiv:1510.06662](#).
- [3] E. Parini and B. Ruf, *On the Moser–Trudinger inequality in fractional Sobolev–Slobodeckij spaces*, J. Analyse Math. 138 (2019), 281–300, [arXiv:1607.07681](#).
- [4] F. Sk, *Remarks on the fractional Moser–Trudinger inequality*, J. Anal. Math. 148 (2022), 323–349, [arXiv:2002.11747](#).